

[86]

Sardar Patel University

**Master of Science Information Technology - II Semester Examinations
PS02CINT53 : Artificial Intelligence**

Date: 30/04/2022

Time: 12.00 pm to 2.00 pm

Marks: 70

Q1 A Choose the most appropriate option.**[8]**

- 1 _____ is a software program that infers the knowledge available in the knowledge base.
 - a. Inference engine
 - b. User interface
 - c. Knowledge base
 - d. None of these
- 2 _____ test determines whether the given program is intelligent or not.
 - a. Covering
 - b. Turing
 - c. Partitioning
 - d. None of these
- 3 _____ provides a powerful framework for reasoning with imprecise and uncertain information.
 - a. ANN
 - b. GA
 - c. Fuzzy logic
 - d. None of these
- 4 _____ is an example of fuzzy logic based system.
 - a. Percentage calculation
 - b. Calculating average
 - c. Calculating EMI for car loan
 - d. Selecting a wash program for washing machine
- 5 ANN knowledge is stored in _____.
 - a. Weights of connections
 - b. Structure
 - c. Transfer function
 - d. None of these
- 6 _____ is an example of linearly separable problems.
 - a. To select one tour destination from many
 - b. To join army or not
 - c. To select a computer from a store
 - d. None of these
- 7 Travelling salesperson problem with large number of cities can be effectively solved with _____.
 - a. ANN
 - b. Expert system
 - c. GA
 - d. None of these
- 8 _____ is an example of encoding strategies in GA.
 - a. Binary
 - b. Decimal
 - c. Tree
 - d. All of these

Q1 B Fill in the blank/True/False or answer in one line (1 mark each). [16]

- 1 List any three characteristics of natural intelligence.
- 2 Knowledge engineer is responsible for the analysis, design and development of the KBS. (True/False): _____.
- 3 Give full form of DIKW.
- 4 Hill climbing search is a heuristic or weak search. (True/False): _____.
- 5 Fuzzy rule based system represents knowledge in terms of fuzzy rules. (True/False): _____.
- 6 The process of transforming the crisp value into linguistic values is called _____.
- 7 List only two types of fuzzy membership function.
- 8 List operations on fuzzy sets.
- 9 Give full form of ANN.
- 10 A function in the nucleus of an artificial neuron is called _____ function.
- 11 _____ is a computer program that simulates the working of natural nervous system.
- 12 All ANN requires data to learn. (True/False): _____.
- 13 Define GA.
- 14 List two applications of GA.
- 15 Give a name of any one genetic operator.
- 16 GA evolution can be stopped when there is no improvement visible in the solution. (True/False): _____.

Q2 Answer in Brief (ANY SEVEN - 2 marks each). [14]

- 1 List any two types of KBS.
- 2 List production system components.
- 3 Give a major difference between a blind search and weak search.
- 4 Define fuzzy set.
- 5 Define fuzzy relations.
- 6 Draw single perceptron with all labels.
- 7 Give two basic categories of learning paradigms for ANN.
- 8 Give an example of mutations on binary encoded individual in GA.
- 9 Give an example of crossover on decimal encoded individuals in GA.

Q3 Write a short note on knowledge based system by providing general structure of KBS. **[8]**

OR

Q3 Write a short note on production system by taking suitable example. **[8]**

Q4 Draw and explain in detail the general structure of fuzzy rule based system. Also give an example of a fuzzy rule. **[8]**

OR

Q4 Design membership functions for low, medium, and high speed of a two wheeler such as motor cycle. **[8]**

Q5 Write a short note on back propagation learning in multi layer perceptron type of ANN. Explain (i) forward pass and (ii) backward pass. **[8]**

OR

Q5 Draw a multilayer ANN for selection of a computer using parameters of your choice. Also provide a fw training data set for the designed network. **[8]**

Q6 Explain the typical GA cycle by giving its diagram in detail. **[8]**

OR

Q6 Solve travelling sales person problem with GA. Take initial population with five cities as follows. **[8]**

$I_0 = \{ R_1 = (2, 3, 5, 1, 4), R_2 = (2, 1, 4, 3, 5) \}$

Create at least 4 valid routes from the above initial population.

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